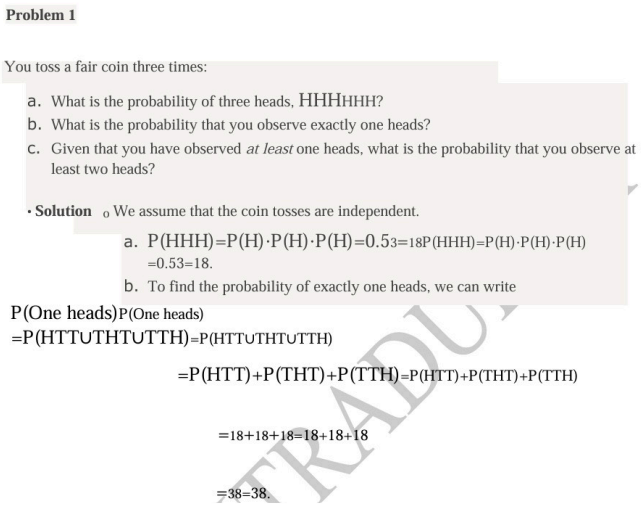


Problem 1

You toss a fair coin three times:

a. What is the probability of three heads, HHHHHH?
b. What is the probability that you observe exactly one heads?
c. Given that you have observed *at least* one heads, what is the probability that you observe at least two heads?

Solution We assume that the coin tosses are independent.

a. $P(HHH) = P(H) \cdot P(H) \cdot P(H) = 0.5^3 = \frac{1}{8}$ $P(HHH) = P(H) \cdot P(H) \cdot P(H) = 0.5 \cdot 3 = \frac{1}{8}$.

b. To find the probability of exactly one heads, we can write

$$P(\text{One heads}) = P(HTT \cup THT \cup TTH) = P(HTT) + P(THT) + P(TTH) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}.$$

explain this concept in student understandable manner

This problem is about **probability of tossing a fair coin three times**. Let us understand it step by step in a very simple way.

Step 1: Understand the Experiment

A **fair coin** has two possible outcomes:

- **H = Head**
- **T = Tail**

Since the coin is tossed **3 times**, each toss is independent.

Each toss has:

$$P(H) = \frac{1}{2}, \qquad P(T) = \frac{1}{2}$$

Step 2: Write the Sample Space

For three tosses, all possible outcomes are:

Outcome
HHH
HHT

Outcome
HTH
HTT
THH
THT
TTH
TTT

There are

$$2^3 = 8$$

possible outcomes.

Since the coin is fair,

Each outcome has probability

$$\frac{1}{8}$$

(a) Probability of getting Three Heads (HHH)

The event is

$$\{HHH\}$$

Only **one outcome** gives three heads.

Using multiplication rule,

$$P(HHH) = P(H) \times P(H) \times P(H)$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{8}$$

Answer

$$\boxed{P(HHH) = \frac{1}{8}}$$

Easy way to remember

Think of three independent tosses.

```
First toss   → H (1/2)
Second toss  → H (1/2)
Third toss   → H (1/2)
```

Multiply all probabilities.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

(b) Probability of getting Exactly One Head

Exactly one head means only **one H** appears.

Possible outcomes are

```
HTT
THT
TTH
```

There are **3 favourable outcomes**.

Each has probability

$$\frac{1}{8}$$

Therefore,

$$P(\text{Exactly one head}) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$

$$= \frac{3}{8}$$

Answer

$$\boxed{\frac{3}{8}}$$

Another method (Combination)

Choose **1 position** for the head out of **3 positions**.

Number of ways

$$\binom{3}{1} = 3$$

Probability of each arrangement

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Therefore,

$$3 \times \frac{1}{8} = \frac{3}{8}$$

(c) Given that at least one head occurred, what is the probability of getting at least two heads?

This is a **Conditional Probability** question.

The formula is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where

- **A = At least two heads**
- **B = At least one head**

Step 1: Find Event B (At least one head)

"At least one head" means

Everything except TTT.

So,

```
HHH
HHT
```

```
HTH
THH
HTT
THT
TTH
```

There are **7 outcomes**.

Thus,

$$P(B) = \frac{7}{8}$$

Step 2: Find Event A (At least two heads)

At least two heads means

Exactly two heads or three heads.

Possible outcomes are

```
HHT
HTH
THH
HHH
```

There are **4 outcomes**.

Thus,

$$P(A) = \frac{4}{8}$$

Notice that every outcome with at least two heads **already has at least one head**, so

$$A \subseteq B$$

Hence,

$$A \cap B = A$$

Step 3: Apply Conditional Probability

$$P(A|B) = \frac{P(A)}{P(B)}$$

$$= \frac{\frac{4}{8}}{\frac{7}{8}}$$

$$= \frac{4}{7}$$

Answer

$$\boxed{\frac{4}{7}}$$

Visual Understanding

Outcome	Heads Count	At least One Head?	At least Two Heads?
HHH	3	✔	✔
HHT	2	✔	✔
HTH	2	✔	✔
THH	2	✔	✔
HTT	1	✔	✗
THT	1	✔	✗
TTH	1	✔	✗
TTT	0	✗	✗

Now ignore **TTT** because the condition says **"at least one head has occurred."**

Remaining outcomes:

```
HHH
HHT
HTH
THH
HTT
THT
TTH
```

Total = **7 outcomes**

Among them, favourable outcomes are

```
HHH
HHT
HTH
THH
```

Favourable = **4 outcomes**

Therefore,

$$P(\text{At least two heads} \mid \text{At least one head}) = \frac{4}{7}$$

Final Answers

Question	Answer
(a) Three heads (HHH)	$\boxed{\frac{1}{8}}$
(b) Exactly one head	$\boxed{\frac{3}{8}}$
(c) At least two heads given at least one head	$\boxed{\frac{4}{7}}$

Key points for students

- **Step 1:** Write the sample space.
- **Step 2:** Count the favorable outcomes.
- **Step 3:** Use the appropriate probability formula.
- **Step 4:** For conditional probability, remember the formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **Step 5:** If all outcomes are equally likely, probability is:

$$\frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}.$$

This counting approach makes probability problems much easier to understand and solve.

c.
d. Given that you have observed at least one heads, what is the probability that
you observe at least two heads? Let A1A1 be the event that you observe at
least one heads, and
A2A2 be the event that you observe at least two heads. Then
A1=S−{TTT}, and P(A1)=78;A1=S−{TTT}, and P(A1)=78;
A2={HHT,HTH,THH,HHH}, and
P(A2)=48.A2={HHT,HTH, THH,HHH}, and

Show more

(c) Given that you have observed at least one head, what is the probability that you observe at least two heads?

This is a **Conditional Probability** problem.

The formula for conditional probability is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where:

- A_1 = Event of getting **at least one head**
- A_2 = Event of getting **at least two heads**

Step 1: Write the Sample Space

When a fair coin is tossed **3 times**, the sample space is:

$$S = \{HHH,\ HHT,\ HTH,\ HTT,\ THH,\ THT,\ TTH,\ TTT\}$$

There are **8 equally likely outcomes**.

Step 2: Define Event A_1 (At Least One Head)

"At least one head" means **all outcomes except TTT**.

$$A_1 = \{HHH,\ HHT,\ HTH,\ HTT,\ THH,\ THT,\ TTH\}$$

Number of outcomes = **7**

Therefore,

$$P(A_1) = \frac{7}{8}$$

Step 3: Define Event A_2 (At Least Two Heads)

"At least two heads" means **either exactly two heads or three heads**.

The outcomes are:

$$A_2 = \{HHT,\ HTH,\ THH,\ HHH\}$$

Number of outcomes = **4**

Therefore,

$$P(A_2) = \frac{4}{8}$$

Step 4: Find $A_2 \cap A_1$

The symbol ∩ means **common outcomes**.

Since every outcome in A_2 already has **at least one head**, all of them are also in A_1.

Therefore,

$$A_2 \cap A_1 = A_2$$

Hence,

$$P(A_2 \cap A_1) = \frac{4}{8}$$

Step 5: Apply the Conditional Probability Formula

$$P(A_2|A_1) = \frac{P(A_2 \cap A_1)}{P(A_1)}$$

Substitute the values:

$$= \frac{\frac{4}{8}}{\frac{7}{8}}$$

To divide fractions, multiply by the reciprocal:

$$= \frac{4}{8} \times \frac{8}{7}$$

The 8's cancel:

$$= \frac{4}{7}$$

Final Answer

$$\boxed{P(A_2|A_1) = \frac{4}{7}}$$

Simple Explanation for Students

Imagine you tossed the coin three times and **someone tells you that at least one head has already appeared**.

This information removes only one impossible outcome:

❌ **TTT**

The remaining possible outcomes are:

Remaining Outcomes	Heads
HHH	3
HHT	2
HTH	2
THH	2
HTT	1
THT	1

Remaining Outcomes	Heads
TTH	1

There are **7 possible outcomes** left.

Among these, the outcomes having **at least two heads** are:

- HHH
- HHT
- HTH
- THH

That is **4 outcomes**.

Therefore,

$$\boxed{\text{Probability} = \frac{\text{Favourable Outcomes}}{\text{Total Remaining Outcomes}} = \frac{4}{7}}$$

Easy Memory Trick

- **At least one head** → Exclude only **TTT** → **7 outcomes remain**.
- **At least two heads** → Count **HHH, HHT, HTH, THH** → **4 outcomes**.
- So the required probability is:

$$\boxed{\frac{4}{7}}$$

This counting method is often the easiest way for beginners to solve conditional probability questions involving coin tosses.

ChatGPT can make mistakes. Check important info.